Network Programming

(SPG 0463)

PBL ASSIGNMENT:

Question 3

(Hemoglobin Tech)

TRADE : CBS 1

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TTO : Mr Amir Hakeem

DATE : 23 August 2025

PBL ASSIGNMENT REPORT GUIDELINES:

1. Evaluation sheet

Problem Statement Rubric

[Click Here]

2. Problem Statement

Reference from PBL1_Q3.docx file.

Problem statement

You and your team mates have been assigned by the company to develop a blood donation system. Upon the discussion, the system will have several options for the users to choose. The options are:

```
********** MAIN MENU **********
    1: New Donor Registration
    2: Search the Blood Type
       2.1: Search by Gender
       2.2: Search by State
    3: Quit
*************************************
Enter your option:
```

You are required to write a program based on 3 options and 2 sub options that your users need to choose. Above is a sample of main menu. You can program your system based on your creativity and marks will be rewarded for that. Below are the details for each option.

3. Introduction

Blood donation is important because it helps save lives by ensuring hospitals have enough blood for patients. However, keeping track of donor information can be difficult without a proper system. A blood donor management system helps solve this issue by allowing users to register new donors, search for blood types, and filter donor records by gender or state.

This project develops a simple console-based program that performs three main tasks: register a new donor, search for donors by blood type, and exit the program. The system also provides sub-options to filter search results by gender or state, which makes it easier and faster to find suitable donors when needed.

The program follows a menu-driven approach, where users select from different options to manage data. Donor information such as name, IC number, age, gender, state, and blood type is stored and organized properly. This makes donor management more efficient while also showing how programming concepts like lists, dictionaries, loops, conditions, and functions can be applied to real-life problems.

Additional knowledge we learned during the making of Hemoglobin Tech

1. Syntax and it uses

```
55          for x, y in b_types.items():
56              print(f"{x}. {y}")
```

```
76          for x, y in states.items():
77              print(f"{x}. {y}")
```

In this line of code it uses the .items() method to return all key-value pairs as tuples from the dictionary **b_types** or **states.** The variable x in both code will return the numbering value while y will return the blood types (for b_types.items) and name of states (for states.items)

```
if gender.lower() == "m":
    return "Male"
elif gender.lower() == "f":
    return "Female"
```

gender.lower() ensures that whether the uses input M/m or F/f, the program still works

```
[x for x in donors if x["type"] == b_type]
```

This is a shorter version of a for loop nested with an if statement. The use of this line creates a new list of donors where the blood type matches the user selected one. The line **x for x in donors** go through each donor in the donor dictionary. The line **if x[“type”] == b_type** basically keeps and store donors where the **“type”** value matches the selected blood type

2. Implementing Dictionary Mapping for the user menu

```
14      states = {
15          "1": "Johor",
16          "2": "Malacca",
```

This code block shows dictionary named “states” is created and it acts as a lookup table where each number that works as the key is directly mapped to a state name assigned which is the value.

```
73      def ch_state():  2 usages
74          while True:
75              print("\nChoose state:")
76              for x, y in states.items():
77                  print(f"{x}. {y}")
78              ch = input("Enter your choice (1-16): ")
79              if ch in states:
80                  return states[ch]
81              else:
82                  print("Invalid choice. Please try again.")
```

Here, .items() method is used to get every key-value pair from the states dictionary. This loop dynamically generates the user menu and it will print each number (x) to its corresponding state (y). By doing this, .items() method will retrieves every number and state pair and allowing the program to automatically generate a complete menu for the user.

Lastly, to process the user’s selection, the code with checks if the input matches a key in the dictionary. If a match is found, the corresponding value is instantly retrieved and returned. By using directory mapping method, we can replaces the need for a long and repetitive chain of codes.

3. Parsing donors.txt for Donor Records

This part shows how the program reads and writes donor data from the donors.txt file instead of hardcoding it inside the code. The file is stored in **JSON format**, which makes it easy to convert between text data and Python dictionaries.

- **Loading Donors**
  The load_donors() function checks if donors.txt exists.
  If yes, it opens the file and uses json.load() to convert the JSON data into a Python list of dictionaries, where each dictionary stores one donor's details.
  If no file exists, it returns an empty list [] so the program can still run without errors.

```python
# Load donors from file
def load_donors():  1 usage
    if os.path.exists("donors.txt"):
        with open("donors.txt", "r") as f:
            return json.load(f)
    return []
```

- **Saving Donors**
  The save_donors(donors) function writes new donor data back into the file.
  It uses json.dump() to save the list into donors.txt.
  The option indent=2 makes the data neat and easy to read.

```python
# Save donors to file
def save_donors(donors):  1 usage
    with open("donors.txt", "w") as f:
        json.dump(donors, f, indent=2)
```

By using this method, the system can **keep donor data permanently** even after the program closes, making it more practical and realistic.

4. System Pseudocode or Flowchart to represent your algorithm.

```
Start
    Repeat:
        Display the Main Menu
        Input Option
        If Option == 1
            Input donor's name
            Input donor's IC number
            Input age
            Input gender
            Input state
        Elif Option == 2
            Choose the blood type
            Display the all donors' with chosen blood type
            Ask the user method they want to filter the blood type with
                If Method ==  Gender
                    Input Male or Female
                    Display the filtered data
                If Method == State
                    Input the State
                    Display the filtered data
        Elif Option == 3
            Display "Exiting Program"
            Exit program
        Else
            Display "Invalid Option"
End
```

5. Briefly explain the flow of the system based on several important screenshots of the program output.

i. Once the code is runned the user will be prompted to choose between 3 choices at the main menu

```
---Hemoglobin Tech---
1. Register as new donor
2. Search Blood Type
3. Exit
Enter option (1-3):
```

ii. If the user choose ‘1’ the user will be ask to enter their credentials

```
---Hemoglobin Tech---
1. Register as new donor
2. Search Blood Type
3. Exit
Enter option (1-3): 1

Donor registeration
Enter your name: Adam bin Mohd Nizam
Enter your IC number: 060307-01-4237
Enter your age: 19
Enter your gender (m/f): m
```

```
Choose state:
1. Johor
2. Malacca
3. Negeri Sembilan
4. Selangor
5. Perak
6. Kedah
7. Penang
8. Perlis
9. Kelantan
10. Terengganu
11. Pahang
12. Sabah
13. Sarawak
14. Kuala Lumpur
15. Putrajaya
16. Labuan
Enter your choice (1-16): 1

Choose blood type:
1. A+
2. A-
3. B+
4. B-
5. AB+
6. AB-
7. O+
8. O-
Enter your choice (1-8): 4
Donor successfully added
```

iii. When the user choose ‘2’ they will be prompted what blood type they want to search by

```
---Hemoglobin Tech---
1. Register as new donor
2. Search Blood Type
3. Exit
Enter option (1-3): 2

Search by blood type
1. A+
2. A-
3. B+
4. B-
5. AB+
6. AB-
7. O+
8. O-
Enter your choice (1-8): 4
```

```
Donors with the blood type B-
Name: Fatimah Zain, IC: 950404-07-6789, Age: 29, Gender: Female, State: Perak, Blood Type: B-
Name: Adam bin Mohd Nizam, IC: 060307-01-4237, Age: 19, Gender: Male, State: Johor, Blood Type: B-
```

iv. Then the user with be asked what filter method they what to use

```
Choose filter method
1. Gender
2. State
3. Exit filter
Choice (1-3):
```

a) If the user choose ‘1’ they have to choose what gender

```
Choose filter method
1. Gender
2. State
3. Exit filter
Choice (1-3): 1
Enter your gender (m/f): m

Filtered Donors:
Name: Adam bin Mohd Nizam, IC: 060307-01-4237, Age: 19, Gender: Male, State: Johor, Blood Type: B-
```

b) If the user choose ‘2’ they have to choose what state

```
Choose filter method
1. Gender
2. State
3. Exit filter
Choice (1-3): 2

Choose state:
1. Johor
2. Malacca
3. Negeri Sembilan
4. Selangor
5. Perak
6. Kedah
7. Penang
8. Perlis
9. Kelantan
10. Terengganu
11. Pahang
12. Sabah
13. Sarawak
14. Kuala Lumpur
15. Putrajaya
16. Labuan
Enter your choice (1-16): 1

Filtered Donors:
Name: Adam bin Mohd Nizam, IC: 060307-01-4237, Age: 19, Gender: Male, State: Johor, Blood Type: B-
```

c) If the user choose ‘3’ they will be redirected to the main menu

```
Choose filter method
1. Gender
2. State
3. Exit filter
Choice (1-3): 3

---Hemoglobin Tech---
1. Register as new donor
2. Search Blood Type
3. Exit
Enter option (1-3):
```

v. When input and choose “3”, the program will halt.

```
---Hemoglobin Tech---
1. Register as new donor
2. Search Blood Type
3. Exit
Enter option (1-3): 3
Exiting Program

Process finished with exit code 0
```

6. Sourcecode

Donors Text File (donors.txt)

[

{"name": "Ali Ahmad", "ic": "900101-10-1234", "age": 35, "gender": "Male", "state": "Selangor", "type": "A+"},

{"name": "Siti Nur", "ic": "920202-14-5678", "age": 32, "gender": "Female", "state": "Johor", "type": "A-"},

{"name": "Rahman Ismail", "ic": "880303-05-3456", "age": 37, "gender": "Male", "state": "Penang", "type": "B+"},

{"name": "Fatimah Zain", "ic": "950404-07-6789", "age": 29, "gender": "Female", "state": "Perak", "type": "B-"},

{"name": "Hafiz Abdullah", "ic": "870505-11-9876", "age": 38, "gender": "Male", "state": "Sabah", "type": "AB+"},

{"name": "Aisyah Karim", "ic": "960606-08-1122", "age": 28, "gender": "Female", "state": "Sarawak", "type": "AB-"},

{"name": "Faizal Rahim", "ic": "930707-06-2233", "age": 31, "gender": "Male", "state": "Kuala Lumpur", "type": "O+"},

{"name": "Nurul Iman", "ic": "970808-12-3344", "age": 27, "gender": "Female", "state": "Malacca", "type": "O-"},

{"name": "Daniel Tan", "ic": "890909-02-4455", "age": 36, "gender": "Male", "state": "Pahang", "type": "AB+"},

{"name": "Mei Ling", "ic": "981010-16-5566", "age": 26, "gender": "Female", "state": "Negeri Sembilan", "type": "A+"}

]

Hemoglobin Python File (hemoglobin.py)

```
import json
import os

# Load donors from file
def load_donors():
    if os.path.exists("donors.txt"):
        with open("donors.txt", "r") as f:
            return json.load(f)
    return []

# Save donors to file
def save_donors(donors):
    with open("donors.txt", "w") as f:
        json.dump(donors, f, indent=2)

# Initialize donors list
donors = load_donors()

states = {
    "1": "Johor",
    "2": "Malacca",
    "3": "Negeri Sembilan",
    "4": "Selangor",
    "5": "Perak",
    "6": "Kedah",
    "7": "Penang",
    "8": "Perlis",
```

```
    "9": "Kelantan",
    "10": "Terengganu",
    "11": "Pahang",
    "12": "Sabah",
    "13": "Sarawak",
    "14": "Kuala Lumpur",
    "15": "Putrajaya",
    "16": "Labuan"
}

b_types = {
    "1": "A+",
    "2": "A-",
    "3": "B+",
    "4": "B-",
    "5": "AB+",
    "6": "AB-",
    "7": "O+",
    "8": "O-"
}

def menu():
    print("\n---Hemoglobin Tech---")
    print("1. Register as new donor")
    print("2. Search Blood Type")
    print("3. Exit")
    op = input("Enter option (1-3): ")
    return op
```

```
def blood_t():
    while True:
        print("\nChoose blood type:")
        for x, y in b_types.items():
            print(f"{x}. {y}")
        ch = input("Enter your choice (1-8): ")
        if ch in b_types:
            return b_types[ch]
        else:
            print("Invalid choice. Please try again.")


def ch_gender():
    while True:
        gender = input("Enter your gender (m/f): ")
        if gender.lower() == "m":
            return "Male"
        elif gender.lower() == "f":
            return "Female"
        else:
            print("Invalid choice. Please enter m or f only")


def ch_state():
    while True:
        print("\nChoose state:")
        for x, y in states.items():
            print(f"{x}. {y}")
        ch = input("Enter your choice (1-16): ")
```

```
        if ch in states:
            return states[ch]
        else:
            print("Invalid choice. Please try again.")

def add():
    print("\nDonor registration")
    donor = {}
    donor["name"] = input("Enter your name: ")
    donor["ic"] = input("Enter your IC number: ")
    donor["age"] = int(input("Enter your age: "))
    donor["gender"] = ch_gender()
    donor["state"] = ch_state()
    donor["type"] = blood_t()
    donors.append(donor)
    save_donors(donors)   # save to donors.txt
    print("Donor successfully added and saved.")

def print_donor(donor):
    print(f"Name: {donor['name']}, IC: {donor['ic']}, Age: {donor['age']}, "
          f"Gender: {donor['gender']}, State: {donor['state']}, Blood Type: {donor['type']}")

def search():
    print("\nSearch by blood type")
    for a, b in b_types.items():
        print(f"{a}. {b}")
    ch = input("Enter your choice (1-8): ")
    if ch not in b_types:
```

```
    print("Invalid blood type")
    return

b_type = b_types[ch]
fil = [x for x in donors if x["type"] == b_type]

if not fil:
    print("No donors found with the blood type.")
    return

print("\nDonors with the blood type", b_type)
for x in fil:
    print_donor(x)

while True:
    print("\nChoose filter method\n1. Gender\n2. State\n3. Exit filter")
    mtd = input("Choice (1-3): ")
    if mtd == "1":
        gender = ch_gender()
        result = [x for x in fil if x["gender"] == gender]
    elif mtd == "2":
        state = ch_state()
        result = [x for x in fil if x["state"] == state]
    elif mtd == "3":
        break
    else:
        print("Invalid option. Please enter 1 until 3")
        continue
```

```
        print("\nFiltered Donors:")
        if result:
            for x in result:
                print_donor(x)
        else:
            print("No matching donor")

while True:
    try:
        option = menu()

        if option == "1":
            add()
        elif option == "2":
            search()
        elif option == "3":
            print("Exiting Program")
            break
        else:
            print("Invalid Option")
    except Exception as e:
        print("An error occurred:", e)
```

7. Conclusion

In conclusion, the blood donor management system provides a simple yet effective way to register new donors and search for blood types. It also includes filtering options such as gender and state, which make it easier to find suitable donors quickly and accurately.

Through its menu-driven design, the program allows donor information to be stored, accessed, and managed in an organized and efficient manner. This ensures that the system is user-friendly while keeping donor records structured and easy to retrieve.

Additionally, this project demonstrates the practical use of basic programming concepts such as data structures, loops, conditions, and functions to solve real-world problems. Overall, the system highlights the importance of technology in managing donor information and supporting the life-saving process of blood donation.

8. References

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- Amir Hakeem, Mr. (n.d.). *Chapter 2: Introduction to Python* [PowerPoint slides]. German-Malaysian Institute (GMI) Virtuelle Lernumgebung (VLU). Course: SPG 0463 Network Programming.

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- GeeksforGeeks. (2025, July 28). *Python Arrays*. In *GeeksforGeeks*. Retrieved August 23, 2025, from https://www.geeksforgeeks.org/python/python-arrays/

Course Name : CYBER SECURITY TECHNOLOGY Course Code : SPG 0463

TTO's Name : MR. AMIR HAKEEM SAHRIR Group : GROUP 2

PROGRAMMING RUBRICS (45)

CRITERIA	MARKS ALLOCATED	MARKS (M)					WEIGHTED MARKS OBTAINED
		1	2	3	4	5	
Ability to apply data manipulation	10	Unable to apply data manipulation in the text file	Able to apply data manipulation in text file but does not apply correctly	Able to apply data manipulation in text file but does not produce correct result.	Able to apply data manipulation in text file and produce partially correct result.	Able to apply data manipulation in text file and produce correct result.	M/5 x 10% =
Ability to use regex	10	Unable to traverse, search, and extract information in text file.	Partially able to traverse, search, and extract information in text file but does not meet requirement.	Able to traverse, search, and extract information in text file but partially meet the requirement.	Able to traverse, search, and extract information in text file but almost meet the requirement.	Able to traverse, search, and extract information in text file and fully meet the requirement.	M/5 x 10% =
Ability to apply required control structure	10	Unable to identify required control structure	Able to identify required control but does not apply correctly	Able to apply required control structure but does not produce correct results	Able to apply required control structure and produce partially correct results	Able to apply required control structure and produce correct results	M/5 x 10% =

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Ability to run/ debug	5	Unable to run program	Able to run program but have logic error	Able to run program correctly without any logic error	Able to run program correctly without any logic error and display inappropriate output	Able to run program correctly without any logic error and display appropriate output	M/5 x 5% =
Ability to produce readable program Comment / Description	5	No documentation	Documentation is simple comment in code	Documentation is simple comments embedded in code with header separating the code	Documentation is simple comments and header that useful in understanding the code	Documentation is well written and clearly explains what the code is accomplishing	M/5 x 5% =
Ability to produce program Indention / Naming Convention	5	Unable to organize the code	The code is poorly organized and very difficult to read	The code is readable only by person who already knows its purpose	The code is easy to read	The code is extremely well organized and easy to follow	M/5 x 5% =
TOTAL	45						**/45**

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WRITTEN REPORT RUBRICS (15)

CRITERIA	MARKS ALLOCATED	MARKS (M)					WEIGHTED MARKS OBTAINED
		1	2	3	4	5	
Ability to analyze problem and identify requirements (input & output)	5	Unable to identify any input and output	Able to identify only one input or output	Able to identify correctly some input and output	Able to identify correctly all input and output	Able to identify correctly all input and output and provide alternative	M/5 x 5% =
Ability to demonstrate design solution	10	Unable to construct flow chart or pseudocode	Able to construct flowchart or pseudocode but mistake on the symbol or sequence	Able to construct flowchart or pseudocode correctly	Able to construct flowchart or pseudocode correctly and use proper elements	Able to construct flowchart or pseudocode correctly, use proper elements and documentation	M/5 x 10% =
TOTAL	15						**/15**

PRESENTATION RUBRIC (10)

CRITERIA	MARKS ALLOCATED	MARKS (M)					WEIGHTED MARKS OBTAINED
		1	2	3	4	5	
Teamwork (Group)	5	Never contribute ideas, do not attend discussion	Rarely contribute ideas, rarely attend discussion	Sometimes contribute ideas, occasionally attend discussion	Usually contribute useful ideas, attend most discussion	Routinely contribute useful ideas, attend all discussion	M/5 x 5% =

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PRESENTATION RUBRIC (10)

Student 01 Name: ABDUL HARIZ BIN MOHD SARNUDIN							
Defense ability (Individual)	5	Not able to Answer questions, not prepared and confidence at all	Able to answer questions but not prepared and confidence	Able to answer questions but with little preparation and confidence	Able to answer Questions well and slightly Confidence and well prepared	Able to all Answer questions very well and confidently. Very well prepared	**MARKS OBTAINED** **/5**

Student 02 Name: MUHAMMAD DANIAL BIN NORHISAM							
Defense ability (Individual)	5	Not able to answer questions, not prepared and confidence at all	Able to answer questions but not prepared and confidence	Able to answer questions but with little preparation and confidence	Able to answer questions well and slightly confidence and well prepared	Able to all answer questions very well and confidently. Very well prepared	**MARKS OBTAINED** **/5**

Student 03 Name: PUTRI NABILA BINTI ZAINOL ABIDDIN							
Defense ability (Individual)	5	Not able to answer questions, not prepared and confidence at all	Able to answer questions but not prepared and confidence	Able to answer questions but with little preparation and confidence	Able to answer questions well and slightly confidence and well prepared	Able to all answer questions very well and confidently. Very well prepared	**MARKS OBTAINED** **/5**

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PBL ASSIGNMENT OVERALL MARKS

	STUDENT 01	STUDENT 02	STUDENT 03
PROGRAMMING (45)			
WRITTEN REPORT (15)			
PRESENTATION (10)			
TOTAL (70)			

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